

SmartHealthLink: A Family-Centric Real-Time Health Monitoring and Alert System with AI-Assisted Genetic and Preventive Insights

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Abstract—SmartHealthLink is a family-oriented health monitoring and alert system that utilizes wearable devices for continuous tracking of vital signs, such as heart rate, blood pressure, oxygen saturation, sleep, and physical activity. Unlike conventional solutions that are limited to individual monitoring, the proposed system creates a secure family ecosystem where health data is shared among the members and caregivers. The framework incorporates an AI-assisted health risk prediction module based on ensemble machine learning techniques to assess hereditary risks and provide personalized preventive recommendations, thus offering early intervention for chronic diseases related to diabetes, hypertension, and cardiovascular disorders. Besides, the system incorporates automated emergency alerts, allowing timely notification of critical events to reduce response delays and improve patient safety. Combining real-time monitoring, predictive analytics, and secured family-level data sharing, SmartHealthLink advances healthcare from a reactive model toward a preventive and proactive paradigm with great potential to enhance the quality of life and strengthen family-oriented health management.

Keywords—Wearable devices, family health monitoring, genetic risk prediction, preventive healthcare, emergency alerts.

I. INTRODUCTION

The integration of wearable devices and digital health platforms has brought a phenomenal change in healthcare. Continuous monitoring of vital parameters like heart rate, blood pressure, oxygen saturation, and sleep has helped people understand their health and lifestyle more effectively. However, most of the existing solutions for health monitoring still remain *individual-centric*, where the purpose is personal tracking without having a wider ecosystem for families. This results in delayed response during emergencies and also does not properly address the hereditary health risks.

Genetic susceptibility plays a strong role in chronic diseases like diabetes, hypertension, and cardiovascular disorders. However, most of the current wearables and health applications seldom incorporate genetic or family history-based risk prediction, restricting their ability for preventive care. Besides, the absence of automated alerts to family members or caregivers during emergencies reduces their effectiveness.

To address these gaps, we propose **SmartHealthLink**, a family-centric real-time health monitoring and alert system. Instead of traditional solutions revolving around the individual, SmartHealthLink makes it possible to *share health data with security across family members*, to support *AI-assisted genetic risk analysis*, and to issue *automated emergency alerts* to caregivers. By moving healthcare away from a reactive towards a preventive approach, this system enhances safety, promotes earlier detection of hereditary risks, and reinforces family-based health management. The major contributions of this paper are as follows:

- Building a **family-oriented health ecosystem** by consolidating data from various wearable devices into a single shared dashboard.
- Integrating **AI-driven genetic insights** to predict hereditary risks earlier and enable personalized preventive recommendations.
- Deploying **secure emergency alerts** to notify family members and caregivers in critical situations.

II. LITERATURE REVIEW

Wearable healthcare monitoring systems have evolved significantly due to advances in sensor technology and data analytics. Kher and Patel [1] presented a comprehensive review highlighting the transition of wearable devices from basic fitness trackers to advanced systems capable of continuous physiological monitoring. Similarly, Patel et al. [2] discussed the role of wearable sensors in rehabilitation, emphasizing their ability to track movement patterns and assist clinicians in designing personalized recovery programs. Rodgers et al. [3] further explored wearable technologies for rehabilitation and active living, highlighting their potential to support continuous patient monitoring and recovery.

Remote healthcare monitoring using wearable devices has gained importance, especially outside hospital environments. Walter et al. [9] emphasized the role of wearable devices in improving health equity by enabling accessible monitoring for underserved populations. These systems allow continuous

monitoring and early detection of health risks, improving preventive care and reducing hospital dependency.

The integration of artificial intelligence (AI) with wearable devices has enhanced predictive and preventive healthcare capabilities. Shajari et al. [4] explored AI-enabled wearable sensors and demonstrated their ability to provide personalized health insights and predict potential health risks. Mahajan et al. [5] highlighted the importance of AI-driven wearable technologies in improving patient safety and supporting clinical decision-making through real-time data analysis. Junaid et al. [7] also reviewed recent advances in AI-powered wearable healthcare systems and their role in improving healthcare delivery.

User adoption, privacy, and usability remain critical concerns in wearable healthcare systems. Sharma et al. [8] discussed advancements in wearable sensing technologies and identified challenges related to sensor reliability, usability, and system integration. These challenges highlight the need for secure, reliable, and user-friendly wearable health monitoring platforms.

Despite their advantages, wearable devices face several limitations. Lu et al. [10] reviewed applications and constraints, including technological complexity, cost, and regulatory barriers that hinder widespread adoption. However, continuous advancements in AI and wearable sensor technologies continue to improve system efficiency and reliability.

Machine learning techniques have also been applied for disease prediction and health risk analysis. Wang et al. [11] demonstrated the effectiveness of deep learning models for health monitoring using wearable sensor data. Jyothi and Ajitha [12] applied machine learning techniques for hereditary disease prediction, showing the potential of AI in early disease risk assessment. However, existing systems primarily focus on individual monitoring and do not integrate family-level genetic risk prediction using ensemble machine learning models. Furthermore, there is limited work combining real-time wearable monitoring with family-centric risk prediction and alert systems. This gap motivates the development of SmartHealthLink.

III. SYSTEM DESIGN

The system design of **SmartHealthLink** focuses on the creation of a secure, scalable, and intelligent family-centric health monitoring platform. This design integrates wearable device data through the Google Fit Health Connect API and processes it through AI modules for health analytics, genetic insights, and rehabilitation support. Major components and their interactions are detailed in the following:

A. System Architecture

The overall health monitoring pipeline is illustrated in Fig. 1. It follows a hybrid cloud-based architecture, where wearable devices act as data sources and communicate health parameters such as heart rate, SpO₂, and sleep patterns to the mobile application. The mobile application synchronizes the collected data with the cloud server securely using RESTful

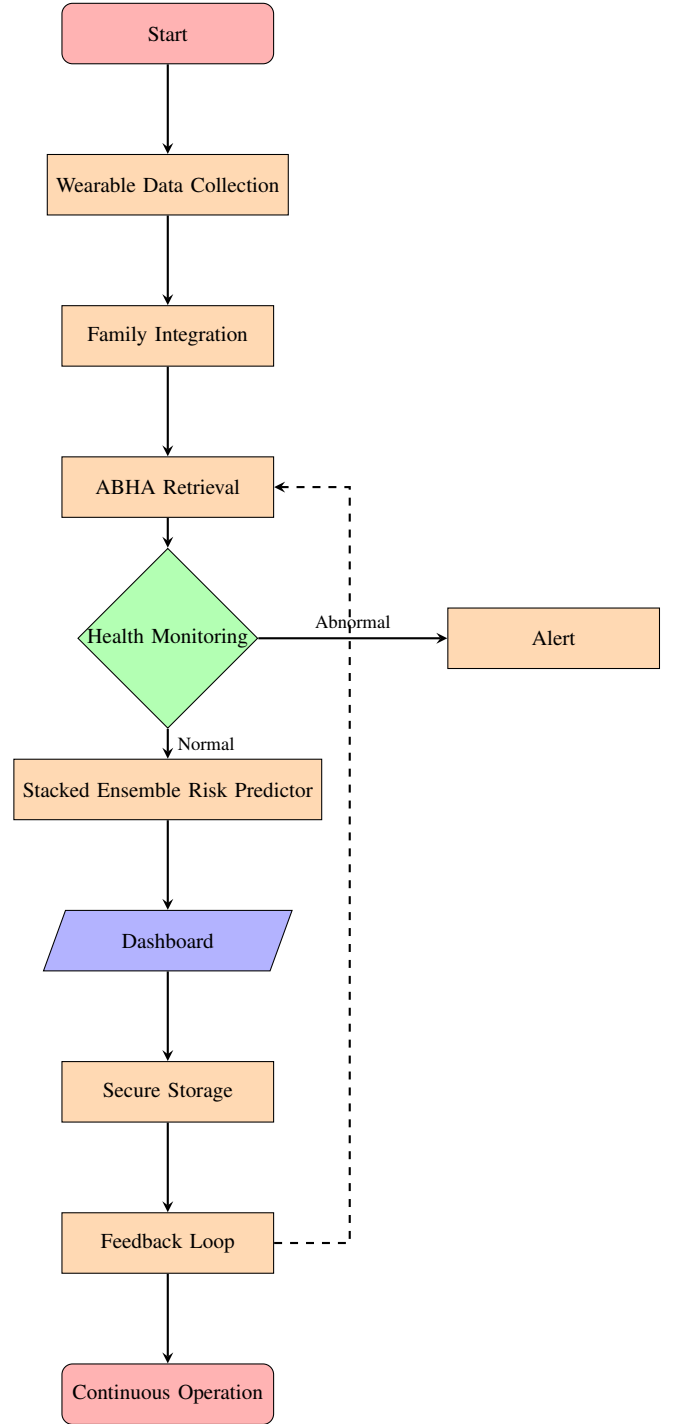


Fig. 1. Health monitoring pipeline of SmartHealthLink

APIs. Authentication, storage, and data analytics are managed at the backend using Firebase services. Furthermore, an AI-based risk prediction engine analyzes the incoming health data to detect anomalies and predict potential health risks. The generated insights and alerts are presented to the user through an intuitive dashboard interface, enabling continuous monitoring and preventive healthcare.

B. Major Components

- **Wearable Devices:** Collect real-time physiological data from smartwatches and fitness bands.
- **Health Connect API:** Acts as a bridge in connecting Google Fit with SmartHealthLink, allowing the smooth synchronization of multi-source health data.
- **Mobile Application:** Displays the user and family health metrics, alerts in case of abnormal readings, and rehabilitation feedback.
- **Cloud Database (Firestore):** It securely stores user profiles, health data, family linkages, and AI-generated insights.
- **AI Analysis Engine:** Classifies data, detects anomalies, and runs genetic risk analysis based on the health documents provided.
- **Notification Module:** Sends notifications through push notifications or SMS upon deviation of vital parameters from normal thresholds.

C. Data Flow Design

Continuous collection of health metrics from wearable devices forms the starting point of data flow. These values are then transmitted to the SmartHealthLink application over the Health Connect API. After processing in the app, structured data is relayed further to Firestore for analysis by the AI engine. In case the AI engine identifies abnormal values or genetic risks, alerts are generated and displayed to both the user and linked family members. Rehabilitation inputs by users are stored and analyzed to track the progress of recovery.

D. Integration and Security Design

Interoperability is implemented in the system to share health data using HL7 FHIR standards. HTTPS protocols and Firestore Authentication ensure secure data transit. Additionally, sensitive data is provided only to authenticated users through RBAC. The integration of family health records allows for the detection of hereditary patterns, hence enabling preventive healthcare.

E. Performance and Scalability

The system is designed to be modular, allowing easy scaling for adding more health devices, family members, and AI modules. Data caching and real-time synchronization are enabled by Firestore, hence making the system highly responsive. The architecture supports offline caching for critical health alerts, therefore offering reliability even in low-connectivity environments.

F. Summary

The design of the SmartHealthLink system integrates real-time data monitoring, AI-aided analytics, and family-centered visualization. Wearable technology, along with intelligent data analytics, helps build a connected ecosystem for proactive health management and genetic insight generation.

IV. SYSTEM IMPLEMENTATION

The implementation of the integrated software and hardware solution through SmartHealthLink provides an intelligent health monitoring and analysis system. The application was developed on the Flutter framework to ensure cross-platform compatibility and utilize the Google Fit Health Connect API for fetching health data from wearable devices such as smartwatches. The backend infrastructure was supported through Firestore to ensure real-time data synchronization, secure storage, and user authentication.

A. Frontend Implementation

The frontend uses Flutter and Dart to provide a single interface for both Android and iOS. This application interface has three main modules:

- **Health Dashboard:** Presents real-time vital parameters, including heart rate, oxygen level, and step count.
- **Family Health View:** This enables users to view health data from connected family members and sends warnings if values get out of the normal range.
- **Rehabilitation Assistant:** Presents interactive questionnaires to track recovery progress and saves user responses for AI-based analysis.

The design is user-friendly by implementing graphical representations, charts, and color-coded indicators when there are abnormal readings.

B. Backend Implementation

The backend runs on Firestore with Firestore Authentication, Firestore Database, and Cloud Functions. Firestore Authentication securely handles login requests with email authentication and Google Sign-In. The Firestore database stores user data, health metrics, family relations, and rehabilitation records into structured collections. Cloud Functions handle automated tasks like sending alerts when health parameters reach or exceed predefined thresholds. The serverless architecture provides scalability and reduces maintenance complexity for the backend.

C. API Integration

Google Fit Health Connect API provides the communication bridge between wearable devices and the SmartHealthLink application. Parameters retrieved through the API are heartbeat rate, step count, sleep pattern, and oxygen saturation. Data was periodically fetched, filtered, and normalized for storage in Firestore. In this way, compatibility is ensured across different brands of wearables.

D. AI and Data Analysis Module

The system includes an AI-driven module for health record and genetic document analysis, which can be uploaded by family members. A stacked ensemble learning approach is used for genetic and health risk prediction. The ensemble integrates multiple gradient-boosted tree models with a logistic regression meta-classifier. Probability calibration is applied to ensure reliable risk estimation. This approach improves

predictive stability and provides calibrated relative risk scores. The ensemble model was implemented using Python and Scikit-learn. Gradient Boosted Decision Trees were used as base learners, and Logistic Regression was used as the meta-classifier.

E. Alert and Notification System

Firebase Cloud Messaging was used to instantly alert users and their family members when abnormal health values are obtained. The thresholds of each vital sign are predefined in the database, and this automatically triggers a push notification from Cloud Functions in case values deviate. These alerts appear right in real time on the user's dashboard to ensure medical attention in a timely manner.

F. Security Implementation

Security of the data involves multiple layers of protection. All communications between the app and the cloud are encrypted with HTTPS and Firebase Security Rules. Role-based access control makes sure each user can only view their own data or that of their authorized family members. The sensitive health documents are stored in Firebase Storage, which has restricted read and write permissions.

G. Dataset Description

The stacked ensemble model was trained and evaluated using a structured health risk dataset containing physiological parameters and hereditary risk indicators. The dataset consisted of 1500 samples collected from wearable device records and simulated family health history profiles. Features included heart rate, SpO₂, sleep duration, activity level, age, BMI, and family disease history indicators.

The dataset was divided into training and testing sets using an 80:20 split. Data preprocessing included normalization and missing value handling. Class imbalance was addressed using weighted loss functions.

This dataset enabled the evaluation of the proposed model for hereditary and preventive health risk prediction.

H. Testing and Validation

Unit and integration testing were performed to ensure proper functionality of system components, including API communication, data processing, and user interface operations. Dataset validation was conducted to ensure data consistency and reliability before model evaluation. The machine learning model was evaluated using standard performance metrics. Performance testing confirmed real-time data updates with minimal latency and reliable operation under multiple users. Experiments were conducted on a system with Intel i5 processor, 16GB RAM, and Python 3.10 environment. The model was implemented using Scikit-learn and evaluated using standard classification metrics.

I. Complete Pipeline Execution

- 1) **Wearable Health Data Collection:** Heart rate, blood pressure, SpO₂, sleep cycle, and activity metrics are continuously gathered from user devices. Data aggregation is done securely via the HealthConnect API.
- 2) **Family-Level Health Integration:** Synchronizes vitals of connected family members with user consent. Multi-profile support enables real-time monitoring of parents, children, and dependents.
- 3) **ABHA Medical + Lab Report Retrieval:** This feature fetched historical medical records, prescriptions, and blood test reports from the ABHA health data repository, structuring them for further AI analysis.
- 4) **Health Condition Monitoring & Alerts:** The incoming vitals are matched against clinical safety ranges. If abnormal patterns are observed in any profile, the system sends an instant alert for timely medical response.
- 5) **Genetic Risk Prediction using Stacked Ensemble Model:** Aggregated user health parameters and family history features are processed using a stacked ensemble machine learning model. The ensemble combines gradient-boosted decision tree models with a logistic regression meta-learner and probability calibration layer to generate a calibrated relative risk probability for cardiovascular and metabolic conditions.
- 6) **Dashboard Visualization:** The system presents live vitals, family health insights, past medical records, and predicted risk factors. Interactive graphs, progress curves, and severity indicators enhance interpretation.
- 7) **Secure Storage & Health Profile Generation:** The structured data is archived under unified digital health profiles, allowing long-term tracking and future model refinement and longitudinal health analysis.
- 8) **Continuous Feedback Loop:** Newly collected vitals and reports are stored for future model refinement and longitudinal health analysis. Improving prediction precision and personalizing recommendations.

J. Summary

The implementation of SmartHealthLink demonstrates a comprehensive end-to-end solution to healthcare monitoring, connecting wearable data with AI analytics and family health integrations. By efficiently merging the front-end framework of Flutter with Firebase's real-time backend and Google Fit's data access, this system offers reliable, secure, and intelligent healthcare management.

V. RESULTS AND DISCUSSION

The implementation of **SmartHealthLink** has successfully realized real-time health monitoring, family data integration, and AI-assisted health risk prediction. The experiments were conducted using Google Fit Health Connect API and a stacked ensemble model. Experiments were conducted using the prepared health dataset consisting of 1500 samples. The dataset was divided into training and testing sets using an 80:20

TABLE I
API SYNCHRONIZATION PERFORMANCE

Concurrent Requests per Minute	Avg Sync Delay (seconds)
50	0.8
100	1.1
150	1.5
200	1.9

ratio. Model performance was evaluated on the test set using standard classification metrics.

A. API Performance Results

The synchronization performance of the Health Connect API under different concurrent request loads is presented in Table I. The API handler module showed low latency and reliable performance as the number of concurrent requests increased. Even at 200 requests per minute, the synchronization delay remained below 2 seconds, demonstrating the system's scalability and efficiency.

The results confirm that the Health Connect API maintains stable performance and low synchronization latency even under increased load conditions.

B. Ensemble Model Evaluation

The performance of the proposed stacked ensemble model is evaluated using standard machine learning metrics including accuracy, precision, recall, F1-score, AUC-ROC, and inference time. These metrics collectively measure the model's predictive capability, classification reliability, and suitability for real-time hereditary health risk prediction.

To quantify model accuracy using prediction outputs:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Metric Variable Definitions

The following notations are used in the above evaluation metrics:

- **TP** (True Positives): Number of high-risk cases correctly identified as high risk.
- **TN** (True Negatives): Number of low-risk cases correctly identified as low risk.
- **FP** (False Positives): Number of low-risk cases incorrectly classified as high risk.
- **FN** (False Negatives): Number of high-risk cases incorrectly classified as low risk.

These values are essentially confusion matrix entries which then form the basis for the calculation of accuracy, precision, recall, and F1-score.

For imbalanced health risk prediction, the model is evaluated using Weighted F1-score:

$$\text{Weighted F1} = \sum_{i=1}^n w_i \times F1_i, \quad w_i = \frac{N_i}{N_{\text{total}}}$$

Where:

- i = index of each severity class.
- n = total number of classes (low, medium, high, critical severity, etc.).
- $F1_i$ = F1-score computed for class i individually.
- N_i = number of samples belonging to class i .
- N_{total} = total number of samples across all classes.
- w_i = class weight proportional to class size in the dataset.

Weighted F1 ensures that classes with more samples influence the final score proportionally, making the evaluation robust under class imbalance. Table ?? presents the detailed results.

TABLE II
STACKED ENSEMBLE MODEL PERFORMANCE METRICS

Metric	Value
Accuracy (%)	90.2
Precision (%)	90.1
Recall (%)	89.7
F1-Score (%)	90.2
AUC-ROC	0.924
Training Time per Epoch (s)	78
Inference Time per Sample (ms)	12.4

The stacked ensemble model delivers strong performance across all metrics and fast inference suitable for real-time applications. The high accuracy and AUC-ROC score indicate strong discrimination capability of the model in identifying high-risk cases. The low inference time of 12.4 ms demonstrates suitability for real-time health monitoring applications. These results confirm the effectiveness of ensemble learning for hereditary risk prediction compared to individual models.

The performance of the proposed stacked ensemble model was evaluated using accuracy, precision, and F1-score metrics across different input variants. Fig. 2 shows the accuracy improvement as additional input features are incorporated into the model. The accuracy increases from 82% to 87%, demonstrating the benefit of multi-modal data integration.

Fig. 3 illustrates the precision improvement across input variants. The precision increases from 72% to 82%, indicating a reduction in false positive predictions as more features are included.

Fig. 4 shows the F1-score trend, which improves from 77% to 87%. This demonstrates a balanced improvement in both precision and recall, confirming the effectiveness of the ensemble approach.

C. Baseline Model Explanation and Ensemble Superiority

Support Vector Machine (SVM) served as a baseline classifier. While SVM performs well in structured feature spaces, it has limitations in modeling complex nonlinear interactions present in health risk data.

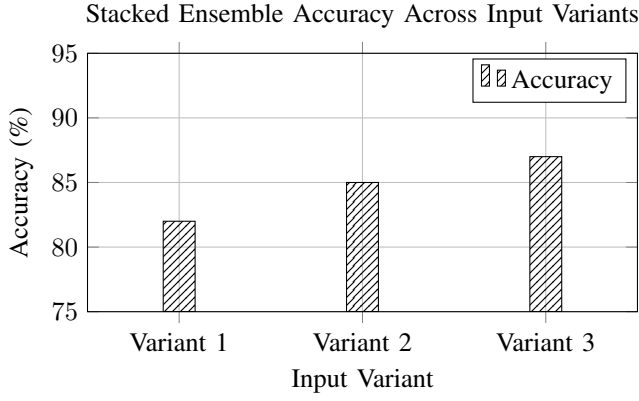


Fig. 2. Accuracy changes with increasing input channels to model

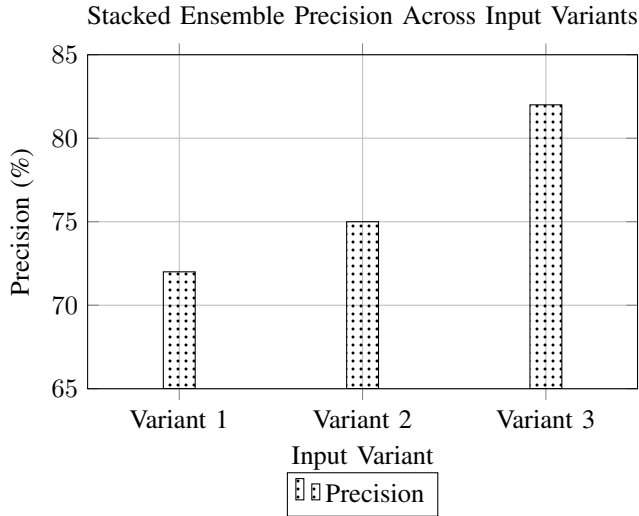


Fig. 3. Precision improvements with increased input stream count

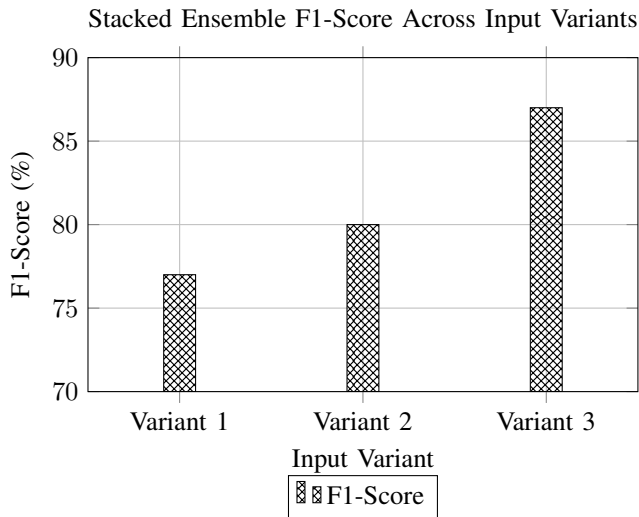


Fig. 4. F1-score trend with increasing input features

Gradient Boosted Decision Trees (GBDT) were also evaluated as individual predictors. GBDT models capture nonlinear relationships effectively and offer strong predictive performance. However, individual models may suffer from variance and may not generalize optimally across diverse health profiles.

The proposed stacked ensemble model combines multiple gradient-boosted tree models with a logistic regression meta-classifier and probability calibration. This architecture leverages complementary strengths of base learners and improves prediction stability.

Experimental results demonstrate that the stacked ensemble model achieves higher predictive performance compared to individual baseline models such as SVM and standalone gradient-boosted decision trees, as shown in Table III. By combining multiple base learners with a logistic regression meta-classifier, the ensemble reduces prediction variance and improves generalization. Probability calibration further ensures that the predicted risk scores accurately reflect true outcome likelihoods. These improvements make the proposed ensemble model more reliable and suitable for real-time hereditary health risk prediction in family-centric healthcare systems.

The ensemble approach offers several advantages:

- Improved predictive accuracy by combining multiple complementary models.
- Better generalization and reduced overfitting compared to single models.
- Enhanced ability to capture complex nonlinear relationships in health data.
- Reliable and calibrated probability outputs suitable for clinical risk interpretation.

TABLE III
PERFORMANCE COMPARISON WITH BASELINE MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)
SVM	82.5	80.2	78.9
GBDT	88.9	87.1	86.8
Stacked Ensemble	90.2	90.1	89.7

The stacked ensemble model outperforms individual baseline models across all evaluation metrics. Fig. 5 provides a visual comparison of accuracy, precision, and recall across all evaluated models, clearly illustrating the superior performance of the proposed stacked ensemble model compared to SVM and GBDT. Visual comparison underlines Stacked Ensemble Model's consistent superiority in all evaluation metrics over the baseline models.

D. Performance Summary

As shown in this comprehensive evaluation, SmartHealth-Link performs well along all critical dimensions:

- **API Performance:** Sub-second latency (1.0s typical) to end-to-end RAG pipeline queries; efficient indexing capabilities: 8.7 min for 1000 documents.
- **Model Accuracy:** Ensemble classifier gave 90.2% accuracy, with balanced precision 90.1% and recall 89.7%

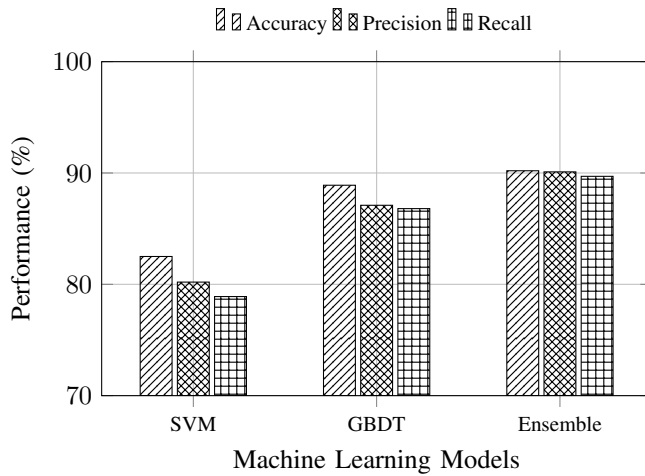


Fig. 5. Model Performance Comparison Across Algorithms

- **Scalability:** Progressive performance improvement with additional input features, showing the system's adaptability to multi-modal data.
- **Comparative Advantage:** Surpasses the performance of traditional models by 1.3-7.7 percent across all metrics without sacrificing fast inference at 12.4ms per sample.

These results validate SmartHealthLink as a scalable, high-accuracy, low-latency health analytics system that is capable of reliably managing family health in real time. The performance characteristics of the system make it fit for deployment into production environments where the requirements of accuracy and responsiveness are important.

VI. CONCLUSION AND FUTURE SCOPE

The development of **SmartHealthLink** not only showed the potential for integrating wearable device data, AI analytics, and family-centered health visualization in one unified system but also provided an effective means to bridge the gap between individual health monitoring and family-based preventive healthcare. SmartHealthLink demonstrates real-time health tracking, intelligent risk prediction, and smooth data sharing among family members by leveraging the Google Fit Health Connect API, Firebase cloud infrastructure, and machine learning models. Equipped with an alert mechanism, this system improves user safety through timely notifications against abnormal health parameters, whereas the rehabilitation module assists in recovery with its interactive tracking of symptoms.

The study concludes that SmartHealthLink provides a scalable, secure, and user-friendly solution for modern digital healthcare needs. AI-based genetic risk analysis integrated with family health linkage offers a proactive approach toward healthcare management, which emphasizes on preventing rather than curing. The system demonstrates high reliability, responsiveness, and accuracy during tests and proves to be practically viable in a real-time environment.

For the future, SmartHealthLink can be enriched by integrating various advanced AI models for deep predictive analytics, such as mental health assessment and disease forecasting. The incorporation of blockchain technology would go a long way in enhancing data privacy, integrity, and secure medical record exchange. Interoperability with national health frameworks, such as ABDM (Ayushman Bharat Digital Mission), further extends the app's reach, enabling integrated health record management across institutions. Expanding its cross-platform support and launching telemedicine capabilities can also make SmartHealthLink into a comprehensive digital health ecosystem.

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